



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 12

Register Number: 1 7 7 6 3 9 2

SECOND REVISION EXAMINATION - 2025

BUSINESS MATHEMATICS AND STATISTICS

Time Allowed : 3.00 Hours]

[Max. Marks : 90

PART - I

20 x 1 = 20

1. Answer all the questions by choosing the correct answer from the given 4 alternatives
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. The rank of the matrix $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$ is
 (a) 0 (b) 1 (c) 2 (d) 3
2. $|A_{n \times n}| = 3$ $|\text{adj}A| = 243$ then the value n is
 (a) 4 (b) 5 (c) 6 (d) 7
3. The value of $\int_2^3 f(5-x)dx - \int_2^3 f(x)dx$ is
 (a) 1 (b) 0 (c) -1 (d) 5
4. $\int 2^x dx$ is
 (a) $2^x \log 2 + c$ (b) $2^x + c$ (c) $\frac{2^x}{\log 2} + c$ (d) $\frac{\log 2}{2^x} + c$
5. If the marginal revenue function of a firm is $MR = e^{-\frac{x}{10}}$, then revenue is
 (a) $-10e^{-\frac{x}{10}}$ (b) $1 - e^{-\frac{x}{10}}$ (c) $10(1 - e^{-\frac{x}{10}})$ (d) $e^{-\frac{x}{10}} + 10$
6. The particular integral of the differential equation is $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 2e^{4x}$
 (a) $\frac{x^2 e^{4x}}{2!}$ (b) $\frac{e^{4x}}{2!}$ (c) $x^2 e^{4x}$ (d) $x e^{4x}$
7. If 'n' is a positive integer $\Delta^n [\Delta^{-n} f(x)]$
 (a) $f(2x)$ (b) $f(x+h)$ (c) $f(x)$ (d) $\Delta f(x)$
8. Lagrange's interpolation formula can be used for
 (a) equal intervals only (b) unequal intervals only
 (c) both equal and unequal intervals (d) none of these.
9. Which one is not an example of random experiment?
 (a) A coin is tossed and the outcome is either a head or a tail
 (b) A six-sided die is rolled
 (c) Some number of persons will be admitted to a hospital emergency room during any hour.
 (d) All medical insurance claims received by a company in a given year.
10. Demand of products per day for three days are 21, 19, 22 units and their respective probabilities are 0.29, 0.40, 0.35. Profit per unit is 0.50 paise then expected profits for three days are
 (a) 21, 19, 22 (b) 21.5, 19.5, 22.5 (c) 0.29, 0.40, 0.35 (d) 3.045, 3.8, 3.85
11. An experiment succeeds twice as often as it fails. The chance that in the next six trials, there shall be at least four successes is
 (a) $240/729$ (b) $489/729$ (c) $496/729$ (d) $251/729$
12. Normal distribution was invented by
 (a) Laplace (b) De-Moivre (c) Gauss (d) all the above
13. In simple random sampling from a population of units, the probability of drawing any unit at the first draw is
 (a) n/N (b) $1/N$ (c) N/n (d) 1

14. The standard error of sample mean is
 (a) $\sigma/\sqrt{2n}$ (b) σ/n (c) $\sigma/\sqrt{n}$ (d) $\sigma^2/\sqrt{n}$
15. A time series consists of
 (a) Five components (b) Four components (c) Three components (d) Two components
16. The LCL for R chart is given by
 (a) $D_2\bar{R}$ (b) $D_2\bar{R}$ (c) $D_3\bar{R}$ (d) $D_3\bar{R}$
17. The Penalty in VAM represents difference between the first _____
 (a) Two largest costs (b) Largest and Smallest costs
 (c) Smallest two costs (d) None of these
18. The purpose of a dummy row or column in an assignment problem is to
 (a) prevent a solution from becoming degenerate (b) balance between total activities and total resources
 (c) provide a means of representing a dummy problem (d) none of the above
19. Solve $xdx + ydy = 0$
 (a) $\frac{x^2}{2} + \frac{y^2}{2} = 0$ (b) $x^2 + y^2 = 0$ (c) $x + y = c$ (d) None of these
20. Find the area bounded by the line $y = 2x + 1$ about the x-axis $x = 0$ to $x = 1$
 (a) 0 (b) 2 (c) 1 (d) 3

PART - II

1. Answer any 7 questions
2. Each question carries 2 marks
3. Question number 30 is compulsory

7 x 2 = 14

21. Integrate the following with respect to $x : \frac{1}{2x^2-9}$
22. If $MR = 20 - 5x + 3x^2$, find total revenue function.
23. Solve: $ydx - xdy = 0$
24. Find $\Delta^2 e^x$
25. In an investment, a man can make a profit of ₹ 5,000 with a probability of 0.62 or a loss of ₹ 8,000 with a probability of 0.38. Find the expected gain.
26. Write any 2 examples for Poisson distribution.
27. What is type I error?
28. The following table gives the number of small-scale units registered with the Directorate of Industries between 1985 and 1991. Show the growth on a trend line by the free hand method.

Year	1985	1986	1987	1988	1989	1990	1991	1992
No. of units (in '000)	10	22	36	62	55	40	34	50

29. What is the Assignment problem?
30. Find the value of x if $\begin{vmatrix} x & 1 \\ 0 & x+1 \end{vmatrix} = 0$.

PART - III

1. Answer any 7 questions
2. Each question carries 3 marks
3. Question number 40 is compulsory

7 x 3 = 21

31. A commodity was produced by using 3 units of labour and 2 units of capital, the total cost is ₹62. If the commodity had been produced by using 4 units of labour and one unit of capital, the cost is ₹56. What is the cost per unit of labour and capital? (Use determinant method).
32. Integrate the following with respect to $x : \log x$
33. The marginal cost function of a commodity is given by $MC = \frac{14000}{\sqrt{7x+4}}$ and the fixed cost is ₹18,000. Find the total cost and average cost.

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34. Solve the differential equation : $(3D^2 + D - 14)y = 13e^{2x}$

35. Find the missing entry in the following table

X	0	1	2	3	4
Y	1	3	9	-	81

36. In a Poisson distribution the first probability term is 0.2725. Find the next Probability term

37. A random sample of 60 observations was drawn from a large population and its standard deviation was found to be 2.5. Calculate the suitable standard error that this sample is taken from a population with standard deviation 3?

38. Construct the cost of living Index number for 2015 on the basis of 2012 from the following data using family budget method.

Commodity	Price		Weights
	2012	2015	
Rice	250	280	10
Wheat	70	85	5
Corn	150	170	6
Oil	25	35	4
Dhal	85	90	3

39. Obtain an initial basic feasible solution to the following transportation problem by using least- cost method.

	D ₁	D ₂	D ₃	Supply
O ₁	9	8	5	25
O ₂	6	8	4	35
O ₃	7	6	9	40
Demand	30	25	45	

40. A coin is tossed four times . Let X be the number of heads . Find the distribution function (or) cumulative distributive function of x .

PART - IV

1. Answer all the questions

7x5=35

2. Each question carries 5 marks

41. a) Two products A and B currently share the market with shares 50% and 50% each respectively. Each week some brand switching takes place. Of those who bought A the previous week, 60% buy it again whereas 40% switch over to B. Of those who bought B the previous week, 80% buy it again where as 20% switch over to A. Find their shares after one week and after two weeks. If the price war continues, when is the equilibrium reached?

(OR)

b) Given below are the data relating to the sales of a product in a district. Fit a straight line trend by the method of least squares and tabulate the trend values.

Year	1995	1996	1997	1998	1999	2000	2001	2002
Sales	6.7	5.3	4.3	6.1	5.6	7.9	5.8	6.1

42. a) Integrate the following with respect to x : $e^{3x} \left[\frac{3x-1}{9x^2} \right]$

(OR)

b) Time taken by a construction company to construct a flyover is a normal variate with mean 400 labour days and standard deviation of 100 labour days. If the company promises to construct the flyover in 450 days or less and agree to pay a penalty of ₹10,000 for each labour day spent in excess of 450. What is the probability that

(i) the company pays a penalty of atleast ₹ 2,00,000? (ii) the company takes at most 500 days to complete the flyover?

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43. a) Solve the following differential equations : $(D^2 + D - 6)y = e^{3x} + e^{-3x}$

(OR)

b) Find the optimal solution for the assignment problem with the following cost matrix.

		Area			
		1	2	3	4
Salesman	P	11	17	8	16
	Q	9	7	12	6
	R	13	16	15	12
	S	14	10	12	11

44. a) The demand and supply function of a commodity are $p_d = 18 - 2x - x^2$ and $p_s = 2x - 3$. Find the consumer's surplus and producer's surplus at equilibrium price.

(OR)

b) Evaluate the following using properties of definite integrals: $\int_0^{\frac{\pi}{2}} \frac{\sin^7 x}{\sin^7 x + \cos^7 x} dx$

45. a) The mean I.Q of a sample of 1600 children was 99. Is it likely that this was a random sample from a population with mean I.Q 100 and standard deviation 15 ? (Test at 5% level of significance).

(OR)

b) Using appropriate interpolation formula find the number of students whose weight is between 60 and 70 from the data given below

Weight in lbs	0-40	40-60	60-80	80-100	100-120
No. of students	250	120	100	70	50

46. a) The probability function of a random variable X is given by

$$p(x) = \begin{cases} \frac{1}{4}, & \text{for } x = -2 \\ \frac{1}{4}, & \text{for } x = 0 \\ \frac{1}{2}, & \text{for } x = 10 \\ 0, & \text{elsewhere} \end{cases}$$

Evaluate the following probabilities. (i) $p(X \leq 0)$ (ii) $p(X < 0)$ (iii) $p(|X| \leq 2)$ (iv) $p(0 \leq X \leq 10)$

(OR)

b) A quality control inspector has taken ten samples of size four packets each from a potato chips company. The contents of the sample are given below, Calculate the control limits for mean and range chart.

Sample number	Observations			
	1	2	3	4
1	12.5	12.3	12.6	12.7
2	12.8	12.4	12.4	12.8
3	12.1	12.6	12.5	12.4
4	12.2	12.6	12.5	12.3
5	12.4	12.5	12.5	12.5
6	12.3	12.4	12.6	12.6
7	12.6	12.7	12.5	12.8
8	12.4	12.3	12.6	12.5
9	12.6	12.5	12.3	12.6
10	12.1	12.7	12.5	12.8

(Given for $n = 5$, $A_2 = 0.58$, $D_3 = 0$ and $D_4 = 2.115$)

47. a) Solve $\frac{dy}{dx} = \frac{x^2 y}{x^2 + y^2}$

(OR)

b) Hospital records show that of patients suffering from a certain disease 80 % die of it. What is the probability that of 6 random selected patients atleast 3 will recover.